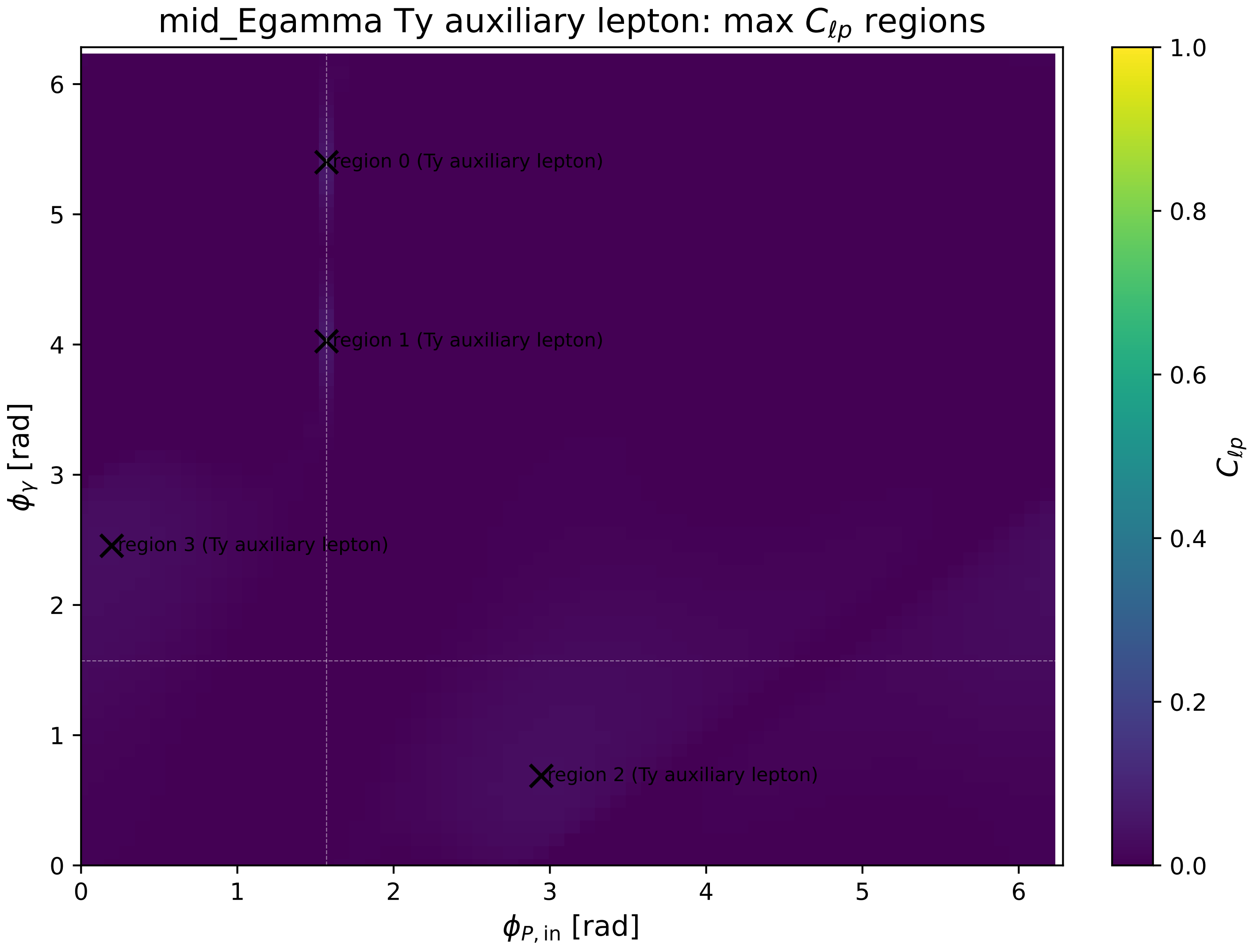
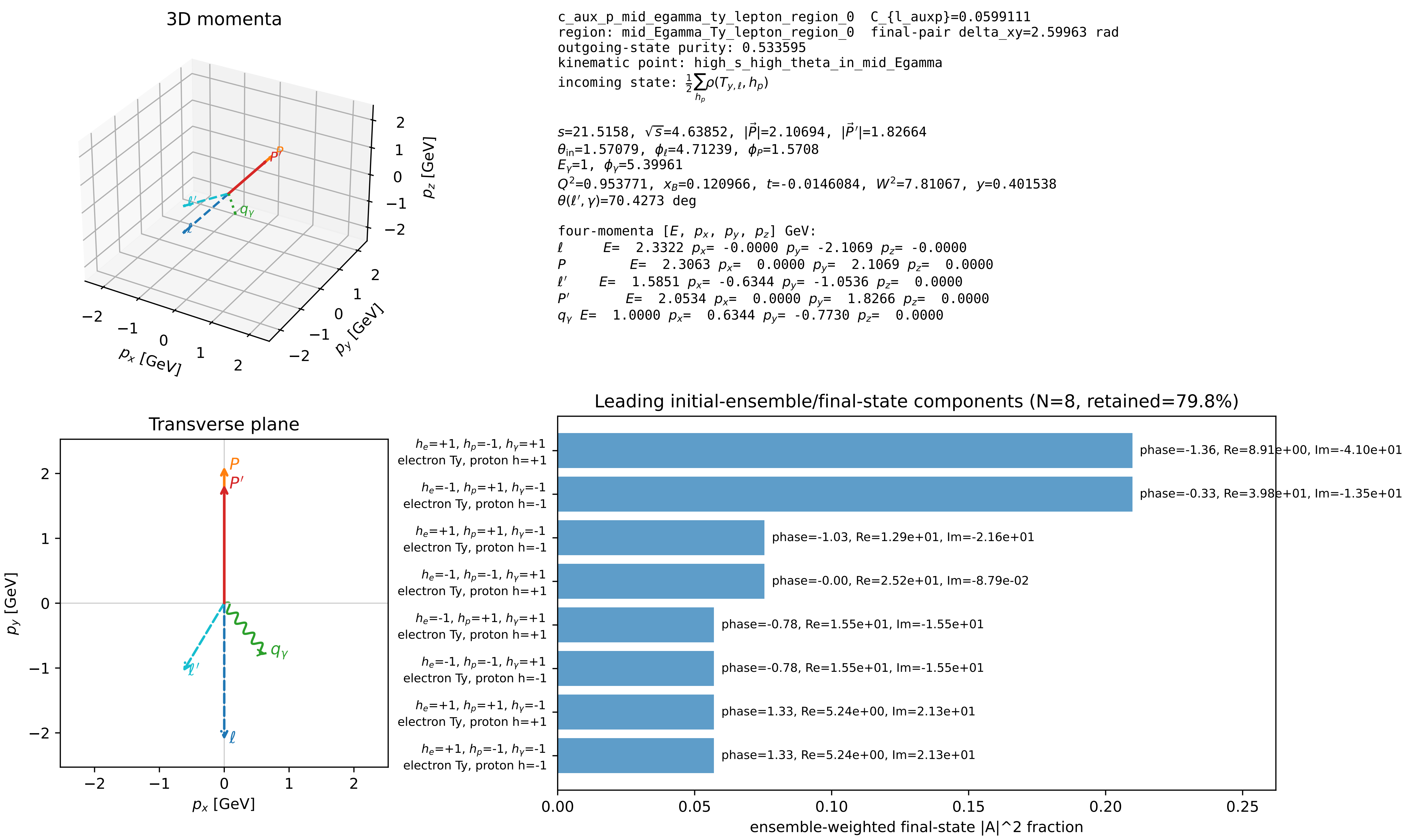
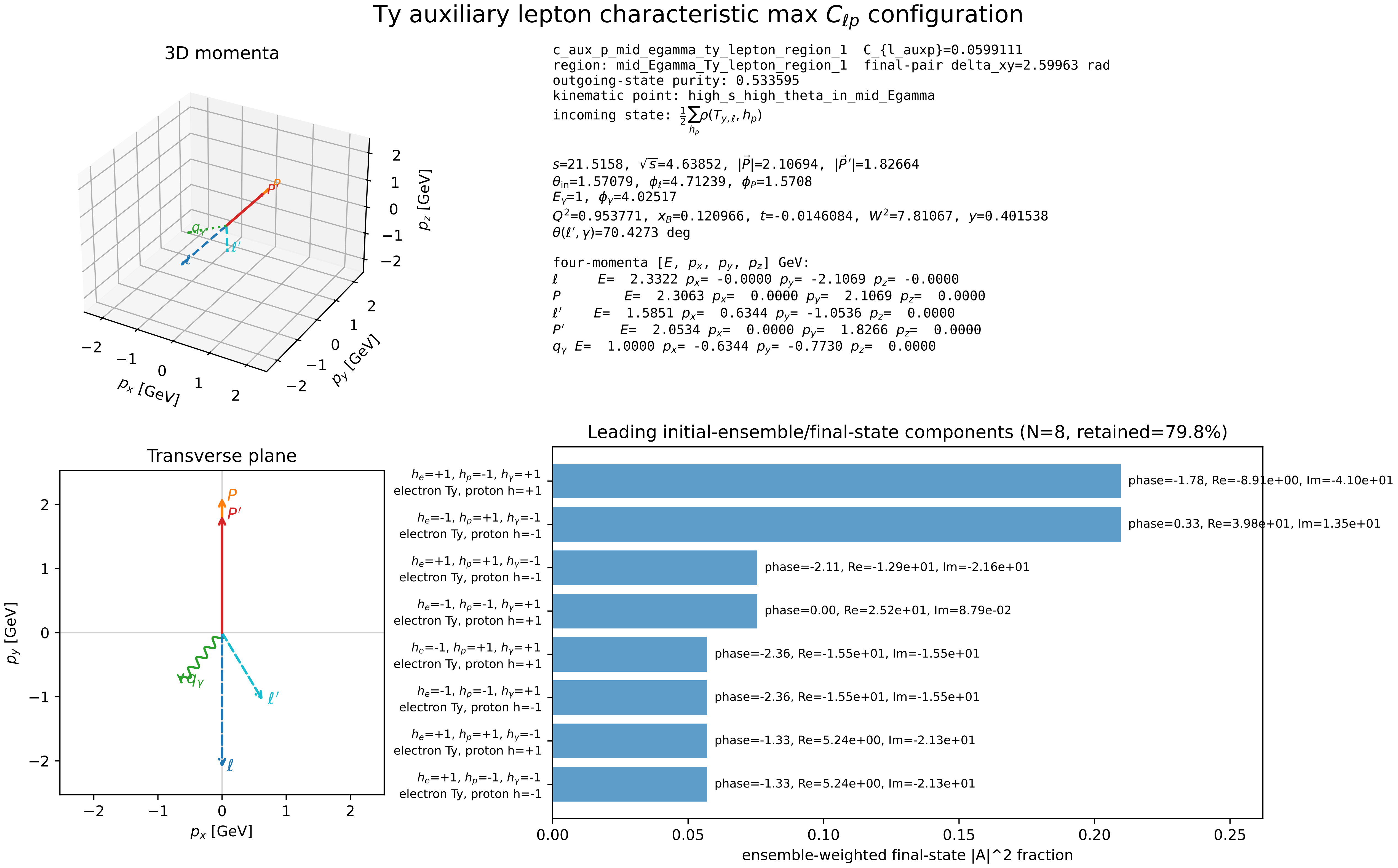


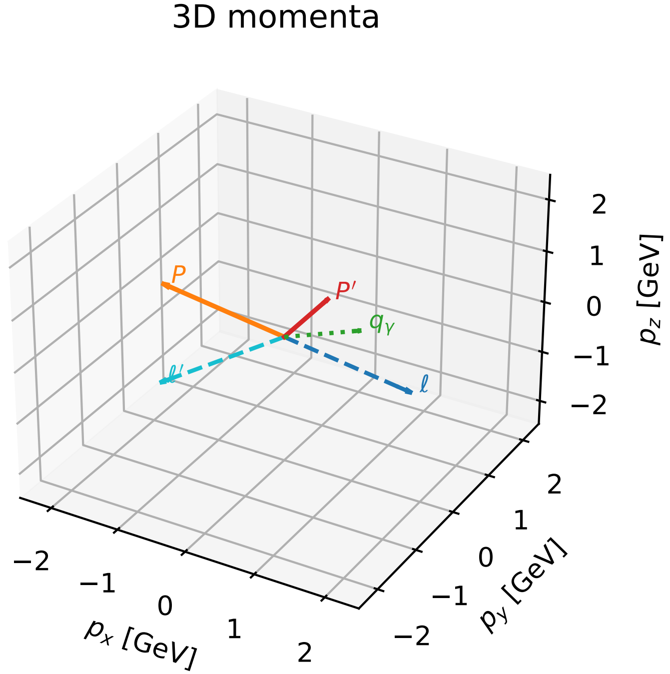
mid_Egamma Ty auxiliary lepton: max $C_{\ell p}$ regions
region 0 (Ty auxiliary lepton)
region 1 (Ty auxiliary lepton)
region 3 (Ty auxiliary lepton)
region 2 (Ty auxiliary lepton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{\ell p}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

Ty auxiliary lepton characteristic max $C_{\ell p}$ configuration




Ty auxiliary lepton characteristic max $C_{\ell p}$ configuration
3D momenta
c_aux_p_mid_egamma_ty_lepton_region_1 C_{l_auxp}=0.0599111
region: mid_Egamma_Ty_lepton_region_1 final-pair delta_xy=2.59963 rad
outgoing-state purity: 0.533595
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell}, h_p)$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.82664
$\theta_{\rm in}$=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=1, ϕ_γ=4.02517
Q^2=0.953771, x_B=0.120966, t=-0.0146084, W^2=7.81067, y=0.401538
$\theta(\ell', \gamma)$=70.4273 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= -0.0000 p_y= -2.1069 p_z= -0.0000
P E= 2.3063 p_x= 0.0000 p_y= 2.1069 p_z= 0.0000
ℓ' E= 1.5851 p_x= 0.6344 p_y= -1.0536 p_z= 0.0000
P' E= 2.0534 p_x= 0.0000 p_y= 1.8266 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.6344 p_y= -0.7730 p_z= 0.0000
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
Leading initial-ensemble/final-state components (N=8, retained=79.8%)
h_e=+1, h_p=-1, h_γ=+1 electron Ty, proton h=+1
phase=-1.78, Re=-8.91e+00, Im=-4.10e+01
h_e=-1, h_p=+1, h_γ=-1 electron Ty, proton h=-1
phase=0.33, Re=3.98e+01, Im=1.35e+01
h_e=+1, h_p=+1, h_γ=-1 electron Ty, proton h=-1
phase=-2.11, Re=-1.29e+01, Im=-2.16e+01
h_e=-1, h_p=-1, h_γ=+1 electron Ty, proton h=+1
phase=0.00, Re=2.52e+01, Im=8.79e-02
h_e=-1, h_p=+1, h_γ=+1 electron Ty, proton h=+1
phase=-2.36, Re=-1.55e+01, Im=-1.55e+01
h_e=-1, h_p=-1, h_γ=+1 electron Ty, proton h=-1
phase=-2.36, Re=-1.55e+01, Im=-1.55e+01
h_e=+1, h_p=+1, h_γ=-1 electron Ty, proton h=+1
phase=-1.33, Re=5.24e+00, Im=-2.13e+01
h_e=+1, h_p=-1, h_γ=-1 electron Ty, proton h=-1
phase=-1.33, Re=5.24e+00, Im=-2.13e+01
ensemble-weighted final-state |A|^2 fraction

Ty auxiliary lepton characteristic max $C_{\ell p}$ configuration



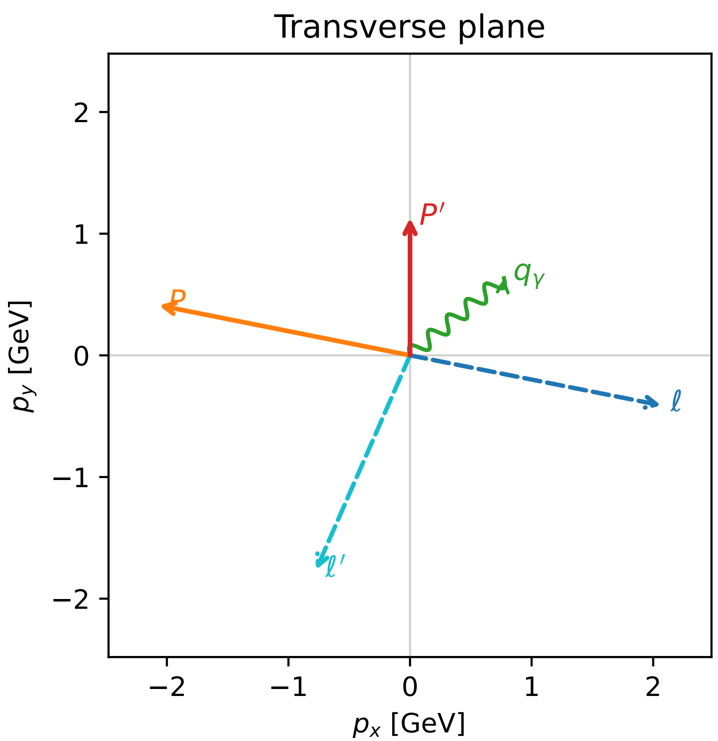

```
c_aux_p_mid_egamma_ty_lepton_region_2  C_{l_auxp}=0.0357188
region: mid_Egamma_Ty_lepton_region_2  final-pair delta_xy=2.7286 rad
outgoing-state purity: 0.500108
kinematic point: high_s_high_theta_in_mid_Egamma
```

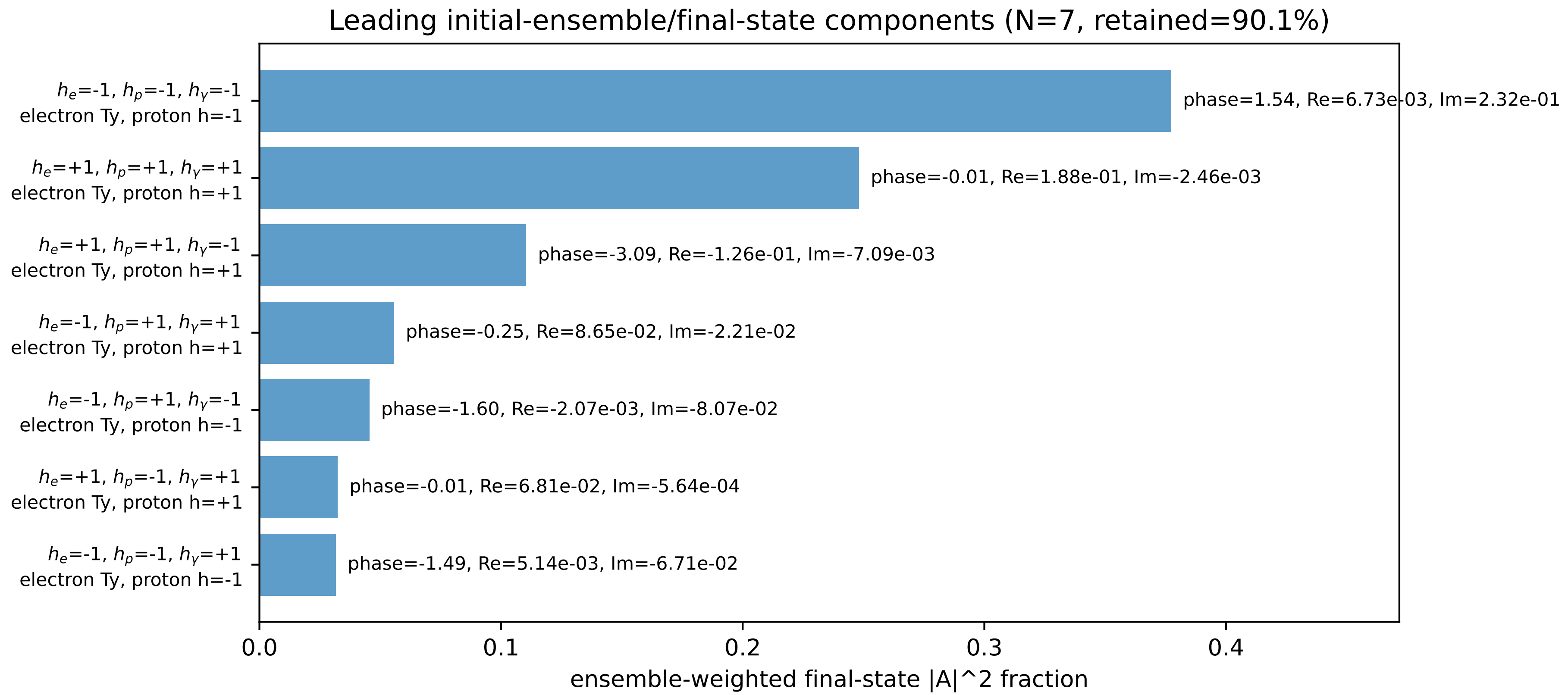
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell}, h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.12971
$\theta_{\rm in}$=1.57079, ϕ_ℓ=6.08684, ϕ_P=2.94524
E_γ=1, ϕ_γ=0.687223
Q^2=9.86708, x_B=0.867783, t=-4.08457, W^2=2.38321, y=0.579061
$\theta(\ell', \gamma)$=153.038 deg

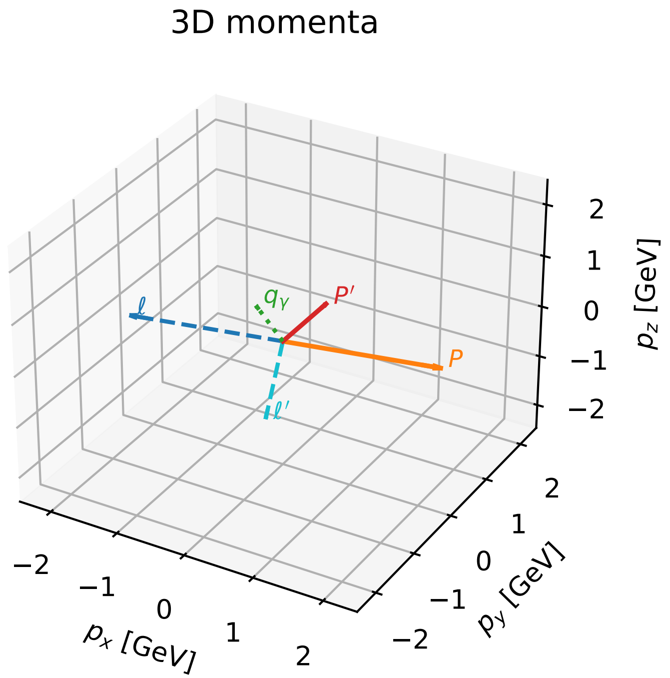
four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 2.3322	p_x= 2.0665	p_y= -0.4110	p_z= -0.0000
P	E= 2.3063	p_x= -2.0665	p_y= 0.4110	p_z= 0.0000
ℓ'	E= 2.1702	p_x= -0.7730	p_y= -1.7641	p_z= 0.0000
P'	E= 1.4684	p_x= 0.0000	p_y= 1.1297	p_z= 0.0000
q_γ	E= 1.0000	p_x= 0.7730	p_y= 0.6344	p_z= 0.0000





Ty auxiliary lepton characteristic max $C_{\ell p}$ configuration




```
c_aux_p_mid_egamma_ty_lepton_region_3  C_{l_auxp}=0.0357188
region: mid_Egamma_Ty_lepton_region_3  final-pair delta_xy=2.7286 rad
outgoing-state purity: 0.500108
kinematic point: high_s_high_theta_in_mid_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell}, h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.12971
θ_{in}=1.57079, ϕ_ℓ=3.33794, ϕ_P=0.19635
E_γ=1, ϕ_γ=2.45437
Q^2=9.86708, x_B=0.867783, t=-4.08457, W^2=2.38321, y=0.579061
$\theta(\ell', \gamma)$=153.038 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= -2.0665 p_y= -0.4110 p_z= -0.0000
P E= 2.3063 p_x= 2.0665 p_y= 0.4110 p_z= 0.0000
ℓ' E= 2.1702 p_x= 0.7730 p_y= -1.7641 p_z= 0.0000
P' E= 1.4684 p_x= 0.0000 p_y= 1.1297 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.7730 p_y= 0.6344 p_z= 0.0000